

Diskrete Mathematik

Woche 3 - HG G 26.5

<https://github.com/tschai-yim/eth-resources>



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Heutige Agenda

1. Teil (45 min)

- 10 min Fragen zu alten Übungen?
- 25 min Theorie-Übersicht mit Beispielen
- 10 min Übungen

10 min Pause

2. Teil: 45 min Übungen

Fragen zu alten Übungen?

Theorie-Übersicht

Kapitel 2.3 - 2.4

Propositional Logic (recap)

$$(A \wedge \neg B) \vee (\neg A \wedge C)$$

- **Motivation:** A formal, mechanical system for reasoning, free of ambiguity.
- Two components:
 - **Syntax:** How you write formulas
 - **Semantics:** What formulas mean

Propositional Logic: Syntax (recap)

- **Propositional Symbols:** Variables for truth values (e.g., A, B, C).
- **Logical Constants:** 0 (false) and 1 (true).
- **Logical Operators:** Combine symbols/formulas.
 - $\neg$ (Negation / NOT)
 - $\wedge$ (Conjunction / AND)
 - $\vee$ (Disjunction / OR, inclusive)
 - $\rightarrow$ (Implication)
 - $\leftrightarrow$ (Two-sided Implication / Equivalence)
- **Formula:** *Correctly* formed expression of symbols and operators
 - Example: $(A \wedge \neg B) \vee C$
 - Precedence: $\neg$ binds strongest (e.g., $\neg A \vee B$ is $(\neg A) \vee B$).

Propositional Logic: Semantics (recap)

- **Truth Tables:** Define operator functions via input-output rules.

A	B	$\neg A$	$A \wedge B$	$A \vee B$	$A \rightarrow B$	$A \leftrightarrow B$
0	0	1	0	0	1	1
0	1	1	0	1	1	0
1	0	0	0	1	0	0
1	1	0	1	1	1	1

- **Apply recursively** to formulas.
 - Reminder: show trick with $(A \wedge \neg B) \vee B$
- **As Functions:** A formula with n symbols induces a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$.
 - Any function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ **can be represented** by a prop. logic formula.

Propositional Logic: Relationships (recap)

Symbol	Type	Input	Output	Context
$F \leftrightarrow G$	Operator	2 Formulas	1 new Formula	In Propositional Logic “world”
$F \equiv G$	Relation	2 Formulas	1 Statement	About Propositional Logic formulas
$S \Leftrightarrow T$	Relation	2 Statements	1 new Statement	“Real World” Mathematics

Symbol	Type	Input	Output	Context
$F \rightarrow G$	Operator	2 Formulas	1 new Formula	In Propositional Logic “world”
$F \models G$	Relation	2 Formulas	1 Statement	About Propositional Logic formulas
$S \Rightarrow T$	Relation	2 Statements	1 new Statement	“Real World” Mathematics

Formulas vs. Statements - Valid or Invalid?

- $A \rightarrow (B \leftrightarrow C)$
 -  **Valid Formula.** Only uses formula operators.
- $(A \equiv B) \rightarrow C$
 -  **Invalid (Type Clash).** $\rightarrow$ expects Formulas, but $(A \equiv B)$ is a Statement.
- $(A \vee B) \Rightarrow A$
 -  **Invalid.** $\Rightarrow$ expects Statements, not Formulas. (Use $\rightarrow$ or $\models$).
- $(F \equiv G) \Leftrightarrow (F \models G \text{ and } G \models F)$
 -  **Valid Statement.** $\Leftrightarrow$ correctly connects two Statements.
- $(F \models G) \leftrightarrow \text{True}$
 -  **Invalid (Mixing Worlds).** $\leftrightarrow$ expects Formulas; $(F \models G)$ is a Statement.
- $\models (A \rightarrow A)$
 -  **Valid Statement.** Shorthand for $\top \models (A \rightarrow A)$ (Tautology).

Formulas vs. Statements - Valid or Invalid?

- $(A \equiv B) \equiv C$
 -  **Invalid (Chaining).** Second $\equiv$ expects a Formula, but $(A \equiv B)$ is a Statement.
- $(A \leftrightarrow B) \models (A \rightarrow B)$
 -  **Valid Statement.** $\models$ correctly compares two Formulas.
- $x \text{ ist gerade} \rightarrow x+2 \text{ ist gerade}$
 -  **Invalid.** $\rightarrow$ is for Formulas. Real-world statements require $\Rightarrow$.

Predicate Logic

$$\exists x(x > 1 \wedge \forall y \forall z((y \cdot z = x) \rightarrow (y = 1 \vee z = 1)))$$

- **Motivation:** Extend propositional logic to cover more things.
- Same two components:
 - **Syntax:** How you write formulas
 - **Semantics:** What formulas mean

Predicate Logic: New Syntax

- **Universe U :** Non-empty set of objects to reason about (e.g., $\mathbb{N}$, $\mathbb{R}$).
- **Constants:** Specific, fixed elements from U (e.g., 0 , 1 , π).
- **Predicates $P(x)$:** Functions mapping elements from U to a truth value, $\{0, 1\}$.
 - Example: $\text{less}(x, y)$ returns 1 if $x < y$.
- **Functions $f(x)$:** Map elements from U to another element in U .
 - Example: $\text{add}(x, y)$ on $\mathbb{N}$ returns $x + y$.
- **Variables:** Placeholders for elements from U (e.g., x, y, z).
- **Quantifiers:** $\forall x(\dots)$ and $\exists x(\dots)$

Def 2.10. A k -ary predicate P on U is a function $U^k \rightarrow \{0, 1\}$.

Predicate Logic: New Semantics

Three layers:

1. **Statements:** English, $\Rightarrow$, $\Leftrightarrow$
2. **Logic** $\{0, 1\}$: Predicate Logic, $\rightarrow$, $\leftrightarrow$
3. **Universe** U : specific interpretation

Example from before:

$$\exists x(G(x, c) \wedge \forall y \forall z (E(m(y, z), x) \rightarrow (E(y, c) \vee E(z, c))))$$

- **Exists a prime in $\mathbb{N}$ (True):** Let $U = \mathbb{N}$, $c = 1$, $G(a, b) \Leftrightarrow a > b$, $E(a, b) \Leftrightarrow a = b$, and $m(a, b) = a \cdot b$.
- **Over $\mathbb{R}$ (False):** Let $U = \mathbb{R}$, using the same symbols.

$\Rightarrow$ not a tautology



Predicate Logic: Quantifier Details

- **Universal Quantifier** ($\forall$): “For all”
- **Existential Quantifier** ($\exists$): “There exists”
- **Nested Quantifiers** (start outside):
 - Above of notation: $\forall x \exists y P(y, x) \equiv \forall x (\exists y (P(y, x)))$
 - “For every $x \in U$ there exists $y \in U$ s.t. $P(y, x) \equiv 1$ ”
- **Variable Scope**: Binds a variable *within its scope*.
 - Example: In $\exists x (x \leq 2) \wedge \exists x (x \geq 3)$, the left x is independent of the right x .

Predicate Logic - Valid or Invalid?

- $\forall x \exists y (P(x, y) \rightarrow Q(y))$
 -  **Valid.** Standard combination of quantifiers and predicates.
- $\forall x (x \wedge y)$
 -  **Type Clash.** x and y are terms. Operator $\wedge$ requires formulas.
- $P(f(x), g(y))$
 -  **Valid.** Functions (f, g) return terms, which are valid args for predicates (P) .
- $\forall x (x + 1)$
 -  **Type Clash.** $x + 1$ is a term. Quantifiers require a formula in their scope.
- $\exists x > 0$
 -  **Syntax Error.** Quantifier must explicitly bind variable (must be $\exists x (x > 0)$).
- $\exists x (P(x)) \wedge \exists x (Q(x))$
 -  **Valid.** Reusing variable names in separate scopes is syntactically legal.

Übungen

<https://github.com/tschai-yim/eth-resources> unter
03_TA/26HS_252-0025-01L_Diskrete_Mathematik/02_Exercises

Offene Fragen & Feedback

- Gibt es noch Fragen zu den heutigen Themen oder den Übungen?
- Habt ihr Feedback zur heutigen Übungsstunde?
- Fragen? Schreibt mir eine E-Mail oder auf WhatsApp.

Schöne Woche und bis nächsten Montag!